Bedfordview Primary School
TERM 2 TEST

Name: MEMO	Grade 5 : __
Subject: Natural Science and Technology	
Examiner: Mrs C. Nash	Date: 2023-06-02
Moderator: Mr J. Nash	Time Allocation: 1hour 45min
	TOTAL 60
INSTRUCTIONS 1. An extra 10 min will be provided for reading. 2. Read all the instructions and questions carefully. 3. Write neatly and legibly. 4. Write in full sentences unless otherwise instructed. 5. Look carefully at the mark allocation when giving your answers. 6. Answer in the spaces provided. 7. This paper consists of 6 pages.	Parent's signature: _____ Comments: _____ _____ _____ _____ _____

1	2	3	4	5	6	7
0 – 29%	30 – 39%	40 – 49%	50 – 59%	60 – 69%	70 – 79%	80 – 100%

	Section A			Section B				Section C				Section D			Section E			
Question	**1**	**2**	**3**	**1**	**2**	**3**	**4**	**1**	**2**	**3**	**4**	**1**	**2**	**3**	**1**	**2**	**Total**	**%**
Allocated mark	2	10	1	6	12	1	3	2	2	2	5	1	1	4	4	4	60	
Learner's mark																		
Moderated mark																		

SECTION A – Definitions / 13

1. In a full sentence explain the meaning of the word "Property" that is used in science. /2

A quality or feature of an object that never changes.

2. Match the term to the correct definitions. Write the correct number next to each letter. /10

Term	Definition
a. Strong	1. how well a material lets electricity or heat travel through it.
b. Hard	2. how much a material can stretch without breaking.
c. Malleable	3. can hold heavy loads without breaking.
d. Plasticity	4. how easily a material can be bent.
e. Tensile strength	5. lustrous.
f. Flexible	6. can be bent or hammered out of shape without cracking or breaking.
g. Conductive	7. solid, firm, and rigid.
h. Shiny	8. how well a material can be shaped.
i. Versatile	9. hard and easily broken.
j. Brittle	10. has many uses.

a) 3 b) 7 c) 6 d) 8 e) 2 f) 4 g) 1 h) 5 i) 10 j) 9

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3. What does it mean when a substance has *solubility*. /1

It is how well a solid or substance will dissolve.

SECTION B – Case Study **/ 22**

A kitchen can either be a place of disaster or a place full of successful cooking.
Read the following case study and answer the questions that follow.

Delicious Delightful Disasters

Chefs in a kitchen are required to perform many different tasks during their working day. The tasks involve preparing food, taking food to the distribution area for waiters to take to customers, washing pots and pans and cleaning cooking equipment.

Chef Charlie was new to the food industry and his first day on the job will be a day he will never forget.

When he arrived at the restaurant's kitchen, he was very excited to get started and immediately put on his white chef's jacket that had his name printed on and washed his hands with soap at the kitchen sink. Chef Charlie took a moment and looked around the enormous kitchen to see where everything was.

The kitchen had lots of natural light coming in through the large windows which made it easier for him. He started right away by taking out the basics that he was going to need for his cooking. Chef Charlie took out his favourite wooden spoon that he had in his back pocket of his jeans; then the shiny cutlery from the kitchen draws and the chopping board. While he was getting himself organized, he did not notice that he had accidently turned the stovetops on as he was reaching for the different size cooking pots and pans that were hanging above him. Chef Charlie had everything neatly placed on the kitchen table, next to a beautiful pot plant and went to the fridges to gather the fresh produce. He was delighted to see the variety of ingredients and spices from which he could choose.

Chef Charlie knew straight away what delicious dishes he was going to cook that day. Getting back to his workstation, Chef Charlie placed all that he needed on the kitchen counter next to the stove and happily skipped across the tiled kitchen floor to close the fridge doors. He got halfway when heard a loud thump which gave him such a fright. Looking around to see just what had happened he could smell something burning. Chef Charlie ran back to the stove and saw that not only the egg carton had caught alight, but also the jam jar had fallen over and broke. "Oh no Charlie!" he said to himself, as he saw the cracked eggs had splattered all over the stovetops and it was an absolute mess. He quickly turned the stove off and went back to the kitchen sink to get a cloth, a few sheets of paper towels, and a bucket with warm, soapy water to clean up the eggy mess. Chef Charlie managed to quickly clean the stovetops and swept the kitchen floor.

While cleaning up he had to rethink of another dish to make as he had lost many of the eggs. On his way back from finally closing the fridge doors he saw a bag of potatoes and had an idea. He tried to open the 25kg plastic bag of potatoes using a sharp fish knife to slice the corner of the bag open. With the knife still in his right hand pointing upwards, he then attempted to move the bag with his left hand and accidently cut himself "not again Charlie!" he said again to himself. Now his chef's jacket is stained with blood, it is everywhere and, his first customers have just arrived…

1. Identify 6 metal objects that Chef Charlie used in the kitchen. /6

Metals
1. Kitchen sink
2. Cutlery
3. Stovetops / Stove
4. Pots & Pans
5. Fridge
6. Fish knife

2. Identify 6 non-metal objects with their correct properties that Chef Charlie used in the kitchen. /12

Non-Metals	List 2 Properties for each object
1. Chef's jacket	1. White, Flexible, Soft, Durable
2. Soap, water	2. Colour of the soap, Cold, Slippery, Solid Water = Transparent, Cold and Wet
3. Windows	3. Transparent, Fragile, Solid, Reflective
4. Wooden Spoon	4. Brown, Plasticity, Flammable
5. Jeans, pants	5. Blue, Flexible, Soft, Durable
6. Kitchen draws	6. Lustrous, Durable, Malleable

Other options of non- metals with the appropriate property:

Chopping board, Kitchen table, Pot plant, Tile floor, Egg carton, Jam jar, Plastic bag, Cloth, Bucket, Paper towels

3. What other object could Chef Charlie have used to open the bag of potatoes?
   Give a reason for your answer. /1

A pair of Scissors.

It is safer to use and can cut through plastic easily.

4.a) What object did Chef Charlie use to sweep the floor with? /1

A broom- Brush & dustpan.

4.b) Give an appropriate property for this object. /1

A broom is strong, rigid, brown, durable, rust resistant.

4.c) What is this object made of? /1

A broom is made up of a wooden (plant material) handle with plant fiber /or plastic bristles.

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SECTION C – Questions and Answers /11

1. What is the definition of rust? /2

The process that occurs when metals are exposed to air or water for a long period of time.

A reddish-brown solid that forms a layer on the outside of the metal.

Rusting usually weakens the metal.

2. Name two metals that can rust easily. /2

Iron

Steel

3. How can you protect a fence from rusting **and** what can you use? /2

You can paint the fence using anti-rust paint. **OR**

Sand the fence with sandpaper and then apply the anti-rust paint.

Look at the table below, which shows the properties of metals before answering the questions.

Item	Metal	Strength	Contains Iron	Malleable	Weight of Metal
Door handle	Brass	Fairly strong	No	Shapes very easily	Heavy
Spade	Steel	Very strong	Yes	Hard to shape	Fairly heavy
Window frame	Aluminium	Very strong	No	Hard to shape	Light
Teaspoon	Stainless steel	Fairly strong	A little	Fairly easy to shape	Fairly light

4.1) Suggest why the teaspoon will not rust. /1

The teaspoon is made of stainless steel and therefore wont rust.

4.2) Explain why a gardener would not choose an aluminium spade to dig with. /1

Aluminium is too light. More effort is needed when digging.

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4.3) Why do factories make door handles from brass? /1

It is fairly strong, wont rust and is easy to shape.

4.4) Why would we not usually make a teaspoon from brass? /1

The teaspoon will be too heavy.

4.5) Suggest why it is better to make window frames out of aluminium and not steel? /1

Aluminium is light in weight and won't rust

SECTION D- Processing of Materials /6

1. Explain what raw materials are? /1

Raw materials are natural sources or materials found in nature and have not gone through any process.

2. What are processed materials? /1

Materials that have been manufactured and changed in some way.

3. Place the following objects into the correct column. /4

Coal Vegetable Soup Leather Shoes Rice

Raw Materials	Processed Materials
Coal	Vegetable soup
Rice	Leather shoes

SECTION E – True and False Questions & Labelling of diagram /8

1. Decide whether the following statements are true or false.
If it is False, change the sentence to make it true. /4

a) The skull is made up of long, thin bones that form a cage-like structure. False

- The ribs and rib-cage are made up of long, thin bones that form a cage-like structure.

b) Our skeleton forms a hard and strong shell structure for the body to support our body's weight and gives us shape.

False

- Our skeleton forms a hard and strong frame structure for the body to support our body's weight and gives us shape.

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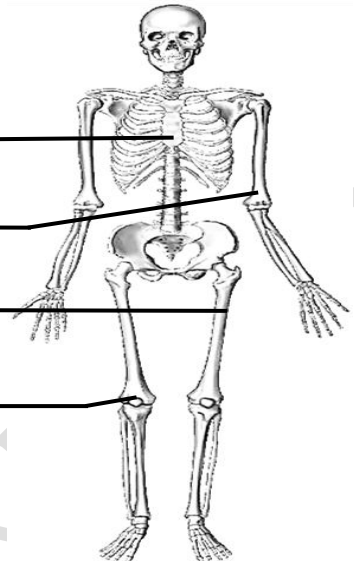
c) The vertebrae are large, flat bones that forms part of the hip-girdle. False

- The hip bones are large, flat bones that forms part of the hip-girdle.

d) The function of the spinal cord is to protect our nervous system. True

2. Fill in the correct labels for the following diagram. /4



a) Sternum

b) Upper limb

c) Lower limb

d) Knee cap/ Knee joint

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